

Robustness, Cascades and Bus Recovery in the Transperth Network

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Abstract

Brief summary of the problem, the modelling approach and the main findings.

1 Introduction

1.1 Background and research aims

A station closure can sever a railway corridor before any further facility fails. Substitute buses may reconnect journeys, but the paths they restore can also concentrate traffic at surviving rail facilities. We investigate these competing effects in the Transperth network, distinguishing infrastructure connectivity, secondary overload failures and the fraction of terminal pairs that remain served. This distinction matters because fewer operational rail facilities need not imply less travel service when another transport layer is available.

Albert et al. (2000) established the distinction between random errors and targeted removal in heterogeneous networks. Their scale-free-network results motivate a comparison of failure strategies here; they do not establish that Transperth has the same degree distribution or resilience. Our audited railway has 86 mapped stations and 85 adjacent segments and is a tree. It therefore lacks alternative rail paths around a removed station. Its inputs represent Monday 5 October 2026, 07:00–09:00 in Australia/Perth, with 271 selected rail trips and 4,324 selected bus trips. These are scheduled services, not measured passenger journeys. Showgrounds remains a mapped station although it has no weekday stop event in this window.

Motter and Lai (2002) showed how a local attack can cause further overload failures when shortest-path loads are recomputed against capacities fixed from the intact network. We use that capacity construction and compare it with a local redistribution scenario in which a failed facility forwards its carried load to surviving neighbours. These rules represent different assumptions about disrupted travel. Their comparison tests whether an apparent tolerance transition depends on the update mechanism. In the bus layer, restored paths can increase surviving rail loads even while improving terminal reachability. Apparent avalanche scaling requires a separate statistical test: following Clauset et al. (2009), a visual slope alone is not evidence for a power law.

Our research questions are: **RQ1**, how do random and targeted removal fragment the railway, and which station characteristics identify bottlenecks? **RQ2**, how do capacity tolerance and load-update rules affect the size and duration of secondary failures? **RQ3**, when do bus backup links improve served travel, and can their rerouting aggravate a rail cascade? **RQ4**, how sensitive are these conclusions to sampling, tolerance-grid resolution and the assumed load/capacity model, and is avalanche scaling statistically supported?

Table 1: Scenarios, parameter ranges and outcome measures.

Scenario	Parameters	Outcome measures
Baseline	values	measures
Variation A	values	measures

1.2 Contribution

Our contribution is a reproducible transport case study that joins an audited GTFS-derived topology to removal, overload and recovery experiments. We expand express-service traversals over mapped adjacent segments rather than treating skipped stops as shortcuts. We separate persistent passenger terminals from closable rail facilities, retain evidence for substitute links, and compare infrastructure damage with service outcomes. Paired scenarios and bootstrap intervals expose uncertainty without treating scheduled departures as observed demand. The capacity law and robustness questions come from existing work; the audited Transperth construction, explicit comparison of load-update assumptions, and evidence-linked recovery analysis are the team’s extension. They are network experiments, not a replication of a taught agent-based model.

2 Model specification

Define the entities, state variables and interactions of the model. The update rule for a generic entity is

$$x_i(t+1) = f(x_i(t), \{x_j(t)\}_{j \in \mathcal{N}_i}; \theta), \quad (1)$$

where $\mathcal{N}_i$ is the neighbourhood of entity i and θ collects the model parameters. Explain each term and justify the form of f , referring to the implementation in `src/` of the main repository.

2.1 Assumptions and initialisation

State the modelling assumptions, boundary cases and initialisation rules, and justify the parameter ranges that are used in the experiments.

3 Experimental design

Describe the experiments used to answer the research question: scenarios, parameter sweeps, random seeds, number of repetitions and outcome measures. Summarise them in a table such as Table 1.

Explain the analysis pipeline that turns raw simulation output into the reported statistics, and point to the notebooks and scripts that implement it.

4 Results

Present the findings in the order of the experiments, referring to each figure and table from the text.

5 Discussion and conclusions

Interpret the results rather than repeating them: what mechanism explains the observed behaviour, and how does it answer the research question?

State the conclusions that the evidence supports, then discuss limitations, uncertainty and the scope of the model. Identify concrete improvements or applications.

References

- Réka Albert, Hawoong Jeong, and Albert-László Barabási. Error and attack tolerance of complex networks. *Nature*, 406:378–382, 2000. doi: 10.1038/35019019. Correction: *Nature* 409, 542 (2001), doi:10.1038/35054111.
- Aaron Clauset, Cosma Rohilla Shalizi, and M. E. J. Newman. Power-law distributions in empirical data. *SIAM Review*, 51(4):661–703, 2009. doi: 10.1137/070710111.
- Adilson E. Motter and Ying-Cheng Lai. Cascade-based attacks on complex networks. *Physical Review E*, 66:065102, 2002. doi: 10.1103/PhysRevE.66.065102.

A Supplementary material

Optional supporting figures, tables or links (for example, to the committed notebooks and results in the main project repository).